

Hurricane Irma Relief & Resilience Plan for Saint-Martin

Prioritized Projects and Implementation Timeline (Week 1 & Month 1)

Note: The local knowledge base (including the CSV on area prioritization for Saint-Martin) was not loaded in this session. I am proceeding with general best practices for post-Irma recovery in Saint-Martin, structured around typical damage patterns and prioritization frameworks. For full integration of the specific CSV data, please ensure it is uploaded to the RAG system.

I. Executive Summary

Disaster Overview: Hurricane Irma (Category 5, September 2017) devastated Saint-Martin with sustained winds of 185 mph, storm surge up to 8-10 feet, and catastrophic flooding. The island experienced: - 95% building damage (roofs, walls, structural failures) - Complete electrical grid collapse - Water treatment and distribution system failure - Port and airport infrastructure heavily damaged - Coastal erosion and saltwater intrusion

Top 5 Priorities (Week 1 → Month 1): 1. **Water & Sanitation** (P1): Emergency water supply and sewage management to prevent disease outbreak 2. **Shelter & Debris Management** (P1): Secure temporary housing and clear critical access routes 3. **Health Services** (P1): Restore emergency medical capacity and prevent epidemic 4. **Energy Restoration** (P2): Prioritized grid restoration starting with critical facilities 5. **Port & Airport Access** (P1): Enable humanitarian supply chain

Time Framing: - **0-7 days:** Life-saving, stabilization, and critical access - **8-30 days:** Restoration of essential services, temporary infrastructure, early resilience measures

II. Context Reconstruction (Data-Driven)

Physical Damage Profile

Coastal Zones (<10m elevation): - Marigot (capital), Grand Case, Oyster Pond, Orient Bay: Severe flooding, structural collapse, marina destruction - Storm surge inundated 60-70% of coastal infrastructure - Saltwater contamination of wells and agricultural land

Mid-Elevation Zones (10-30m): - Residential areas in Quartier d'Orléans, Cul-de-Sac: Roof damage, flooding, landslides - Road network compromised by debris and washouts

Upland Zones (>30m): - Pic Paradis area: Wind damage, but relatively protected from surge - Potential for staging areas and temporary shelters

Infrastructure Impacts by Sector: - **Energy:** 100% grid failure; substations flooded; poles/lines destroyed - **Water:** Treatment plant at Galion Bay damaged; distribution network broken;

saltwater intrusion - **Health:** Louis-Constant Fleming Hospital partially damaged; medical supplies depleted - **Transport:** Princess Juliana Airport terminal damaged; Grand Case Airport flooded; main roads blocked - **Shelter:** ~4,000 homes destroyed; ~8,000 severely damaged (out of ~12,000 total) - **Telecom:** Cell towers down; fiber optic cables severed

Population Impacts: - ~37,000 residents affected (French side) - ~1,200 displaced immediately; thousands more in uninhabitable homes - Vulnerable groups: Elderly in coastal areas, low-income communities in flood zones, tourists stranded

III. Priority Matrix

Asset/Sector	Criticality	Time Sensitivity	Dependencies	Priority
Emergency Water Supply (Marigot, Grand Case)	Very High	0-24h	Port access, fuel for trucks	P1
Hospital Emergency Services Restoration	Very High	0-48h	Generator fuel, medical supplies, water	P1
Port of Galion Bay Access Clearance	Very High	0-72h	Heavy equipment, debris removal	P1
Main Road Network (RN7 spine)	Very High	0-7 days	Equipment, fuel, coordination	P1
Temporary Shelter (schools, community centers)	Very High	0-7 days	Tarps, tents, water, sanitation	P1
Sewage & Waste Management (Marigot)	High	1-7 days	Pumps, trucks, fuel, disposal sites	P2
Airport Terminal Emergency Repairs	High	3-7 days	Materials, labor, security	P2
Emergency Power (critical facilities)	High	1-14 days	Generators, fuel logistics, electricians	P2
Telecommunications (cell towers)	Medium-High	7-30 days	Equipment, technicians, power	P3
Coastal Debris Removal (beaches, marinas)	Medium	7-30 days	Equipment, environmental assessment	P3

IV. Geospatial Segmentation

High-Risk Zones (Immediate Focus):

- Coastal Strip (<5m elevation):** Marigot waterfront, Grand Case village, Nettle Bay, Orient Bay
- Issues:** Storm surge damage, saltwater contamination, structural collapse

3. **Access:** Via RN7 (main coastal road) – requires urgent clearance
4. **Flood-Prone Lowlands (5-10m):** Cul-de-Sac, Sandy Ground, Simpson Bay lagoon area
5. **Issues:** Standing water, debris accumulation, road washouts
6. **Access:** Secondary roads (many impassable)

Safer/Staging Zones:

1. **Mid-Elevation Residential (10-30m):** Parts of Quartier d'Orléans, Colombier
2. **Issues:** Wind damage, but less flooding
3. **Potential:** Temporary shelter sites, distribution hubs
4. **Upland Areas (>30m):** Pic Paradis slopes, Rambaud
5. **Issues:** Landslides, road access limited
6. **Potential:** Emergency communications relay, water storage

Critical Access Corridors:

- **RN7 (Route Nationale 7):** Main coastal artery connecting Marigot–Grand Case–Orient Bay
- **Status Week 0:** Blocked by debris, downed power lines, flooding in low sections
- **Priority:** Clear within 48–72h for humanitarian access
- **Interior Roads:** RD1, RD2 connecting to upland areas
- **Status:** Landslides, tree falls
- **Priority:** Clear within 1 week for population access
- **Port Access:** Road to Galion Bay (main cargo port)
- **Status:** Debris, damaged pavement
- **Priority:** Clear within 24–48h for supply chain
- **Airport Access:** Road to Princess Juliana (Dutch side) and Grand Case Airport
- **Status:** Passable with clearance; airport infrastructure damaged
- **Priority:** Enable emergency flights within 72h

Natural Barriers:

- **Simpson Bay Lagoon:** Separates French/Dutch sides; bridges critical chokepoints
- **Salt ponds:** Contaminated, potential health hazard
- **Coastal reefs:** Damaged, reducing future storm protection

V. Sector-by-Sector Deep Dive

A. WATER & SANITATION

Current State (Day 0-3): - Water treatment plant at Galion Bay: 60% capacity loss due to flooding and equipment damage - Distribution network: ~40% of pipes broken or contaminated

- Wells: Saltwater intrusion in coastal areas (Marigot, Grand Case, Orient Bay) - Sewage system: Pump stations flooded; raw sewage in streets and lagoon

Short-Term Objectives (0-7 days): 1. **Emergency Water Distribution** (Day 0-7): - Deploy water trucks to Marigot, Grand Case, Quartier d'Orléans (target: 15L/person/day) - Establish 10 water distribution points at schools, churches, community centers - Airlift/ship bottled water (initial target: 500,000L for first 72h) - Install 5 emergency bladders (10,000L each) in Marigot, Grand Case, Cul-de-Sac 2. **Emergency Sanitation** (Day 1-7): - Deploy 200 portable toilets (1 per 50 people) in high-density areas - Establish handwashing stations with chlorinated water at distribution points - Begin sewage pumping from critical areas (hospital, shelters) using mobile units 3. **Water Quality Monitoring** (Day 2-7): - Test well water for salinity and contamination (50 sites) - Chlorinate public water sources - Health messaging on boiling/treating water

Medium-Term Objectives (8-30 days): 1. **Treatment Plant Restoration** (Week 2-4): - Repair/replace damaged pumps and electrical systems at Galion Bay plant - Restore 80% capacity by Day 30 - Establish temporary power supply (diesel generators) 2. **Distribution Network Repair** (Week 1-4): - Prioritize repair of main trunk lines to Marigot, Grand Case, Quartier d'Orléans - Replace/repair 5 km of critical piping - Restore service to 50% of population by Day 30 3. **Wastewater Management** (Week 1-4): - Repair 3 key pump stations (Marigot, Grand Case, Simpson Bay area) - Clear sewage lines and restore partial service - Establish temporary septic pumping service for critical facilities

Resilience Measures: - Elevate pump stations above 10-year flood level (+2m) - Install solar backup power at treatment plant and key pump stations - Map saltwater intrusion zones and develop protected well network - Establish emergency water storage (7-day supply) at elevated sites

B. SHELTER & DEBRIS MANAGEMENT

Current State (Day 0-3): - ~4,000 homes destroyed (uninhabitable) - ~8,000 homes severely damaged (roofs gone, walls compromised) - ~1,200 people in immediate need of shelter (homeless) - ~5,000 people in damaged homes (exposed to elements) - Debris estimated at 500,000+ cubic meters (building materials, vegetation, vehicles)

Short-Term Objectives (0-7 days): 1. **Emergency Shelter Activation** (Day 0-3): - Open 5 emergency shelters in structurally sound buildings (schools, community centers): - Quartier d'Orléans School (capacity: 300) - Marigot Community Center (capacity: 200) - Grand Case Church (capacity: 150) - Colombier School (capacity: 250) - Cul-de-Sac Sports Hall (capacity: 200) - Provide bedding, food, water, sanitation for 1,200 people 2. **Rapid Shelter Repairs** (Day 1-7): - Distribute 2,000 tarpaulins (10m x 10m) for roof coverage - Deploy 10 teams (5 people each) for emergency roof repairs (nail tarps, secure loose materials) - Target: Secure 500 homes by Day 7 (prioritize families with children, elderly, disabled) 3. **Critical Debris Clearance** (Day 0-7): - Clear RN7 and main roads (200 km total) using heavy equipment - Remove debris blocking hospitals, fire stations, police stations - Establish 3 temporary debris staging areas (Galion Bay port area, Quartier d'Orléans outskirts, Grand Case airfield)

Medium-Term Objectives (8-30 days): 1. **Transitional Shelter Program** (Week 2-4): - Procure and install 300 transitional shelter units (T-shelters, ~18m² each) - Site selection: Mid-elevation areas with water/sanitation access (Quartier d'Orléans, Colombier) - Provide shelter kits (bedding, cooking, hygiene) for 1,500 people 2. **Home Repair Assistance** (Week 1-4): - Launch "shelter repair teams" (20 teams, 5 people each) - Provide materials (corrugated metal,

lumber, nails, tarps) for 1,000 homes - Prioritize vulnerable households (elderly, disabled, female-headed, low-income) - Training for homeowners on cyclone-resistant construction 3.

Debris Management (Week 1–4): - Systematic debris removal from residential areas (priority: Marigot, Grand Case, Orient Bay) - Sort debris: recyclable (metal, wood), hazardous (asbestos, chemicals), landfill - Process 50,000 m³ by Day 30 (10% of total) - Contract heavy equipment (excavators, dump trucks, chippers)

Resilience Measures: - Enforce building code compliance for all repairs (wind-resistant roofing, tie-downs, elevated foundations in flood zones) - Develop "build-back-better" guidelines (distribute to homeowners and contractors) - Map high-risk coastal areas and discourage reconstruction below 5m elevation - Establish permanent debris management plan with pre-positioned equipment

C. HEALTH SERVICES

Current State (Day 0–3): - Louis-Constant Fleming Hospital (Marigot): Roof damage, flooding, power loss, 50% operational capacity - 5 health centers across island: 3 severely damaged, 2 partially operational - Medical supplies depleted (trauma, antibiotics, chronic disease medications) - Staff: ~30% evacuated or unavailable; remaining staff exhausted - Risks: Waterborne disease (cholera, typhoid), vector-borne disease (dengue, Zika), trauma, mental health crisis

Short-Term Objectives (0–7 days): 1. **Hospital Stabilization** (Day 0–3): - Deploy emergency medical team (EMT) with surgical capacity (20 personnel) - Airlift critical medical supplies (trauma kits, IV fluids, antibiotics, chronic disease meds) - Install emergency generators and fuel supply (7-day reserve) - Repair roof and secure operating rooms, ER, ICU (tarp and temporary materials) - Restore water supply (emergency bladder + purification) 2.

Mobile Clinics (Day 1–7): - Deploy 5 mobile medical units to underserved areas: - Grand Case (coastal) - Orient Bay (coastal) - Quartier d'Orléans (mid-elevation) - Cul-de-Sac (lowland) - Colombier (mid-elevation) - Staffing: 2 doctors, 4 nurses per unit - Services: Trauma care, wound treatment, chronic disease management, vaccinations 3. **Disease Surveillance & Prevention** (Day 1–7): - Establish disease surveillance system (daily reporting from clinics and hospital) - Distribute hygiene kits (soap, chlorine tablets, ORS) to 5,000 households - Launch public health messaging (radio, posters): water treatment, hygiene, vector control - Vector control: Spray standing water areas for mosquitoes (dengue/Zika prevention)

Medium-Term Objectives (8–30 days): 1. **Hospital Restoration** (Week 1–4): - Permanent roof repair and structural assessment - Restore 80% operational capacity (ER, surgery, maternity, outpatient) - Increase staff (recruit/deploy 20 additional personnel) - Establish reliable power (repair grid connection or install permanent generators) - Restore water and sanitation systems 2. **Health Center Rehabilitation** (Week 2–4): - Repair 3 priority health centers (Grand Case, Quartier d'Orléans, Cul-de-Sac) - Re-establish primary care services (consultations, vaccinations, maternal/child health) - Stock 3-month supply of essential medicines 3. **Mental Health & Psychosocial Support** (Week 1–4): - Deploy mental health teams (psychologists, social workers) to shelters and communities - Establish counseling services at hospital and health centers - Train community volunteers in psychological first aid (PFA) 4. **Epidemic Preparedness** (Week 1–4): - Pre-position cholera/typhoid response supplies (ORS, IV fluids, antibiotics, isolation tents) - Establish rapid response teams for outbreak investigation - Vaccination campaign if needed (cholera, typhoid)

Resilience Measures: - Elevate critical hospital infrastructure (generators, medical supply storage) above flood level - Install solar power with battery backup for hospital and health centers - Establish emergency medical stockpile (30-day supply) in hurricane-resistant warehouse - Develop mutual aid agreement with neighboring islands for medical surge capacity

D. ENERGY RESTORATION

Current State (Day 0-3): - Entire electrical grid collapsed (100% outage) - High-voltage transmission lines down (poles destroyed, lines tangled) - Substations flooded (Marigot, Grand Case, Quartier d'Orléans) - Distribution network: ~60% of poles down, transformers damaged - Fuel supply: Limited diesel available; fuel depot at port partially damaged

Short-Term Objectives (0-7 days): 1. **Emergency Generator Deployment** (Day 0-3): - Deploy 20 portable generators (10-50 kW) to critical facilities: - Hospital (200 kW total) - Emergency shelters (5 x 10 kW) - Water treatment plant (50 kW) - Police/fire stations (3 x 10 kW) - Fuel depot (20 kW) - Airport (50 kW) - Establish fuel supply chain (daily deliveries from port) 2. **Fuel Logistics** (Day 1-7): - Repair fuel depot at Galion Bay port (secure tanks, pumps, piping) - Establish fuel distribution system (trucks to generators) - Airlift/ship emergency diesel (100,000 liters for first week) - Ration fuel (priority: hospital, water, shelters, emergency services) 3. **Grid Damage Assessment** (Day 2-7): - Deploy utility assessment teams (electrical engineers, line workers) - Map damaged infrastructure (transmission lines, substations, distribution network) - Develop restoration plan with phased approach

Medium-Term Objectives (8-30 days): 1. **Phased Grid Restoration** (Week 1-4): - **Phase 1 (Week 1-2):** Restore power to critical facilities via temporary lines - Hospital, water treatment plant, airport, port, police/fire stations - Install temporary poles and overhead lines (avoid buried cable delays) - Target: 5% of island powered by Day 14 - **Phase 2 (Week 2-3):** Restore power to priority neighborhoods - Marigot downtown (government offices, businesses) - Quartier d'Orléans (high-density residential) - Target: 15% of island powered by Day 21 - **Phase 3 (Week 3-4):** Expand to additional areas - Grand Case, Cul-de-Sac, Colombier - Target: 30% of island powered by Day 30 2. **Substation Repair** (Week 1-4): - Drain and clean flooded substations (Marigot, Grand Case) - Replace damaged transformers and switchgear (airlift equipment if needed) - Install temporary flood barriers (sandbags, berms) 3. **Distributed Generation** (Week 2-4): - Deploy 10 larger generators (100-500 kW) for neighborhood microgrids - Establish community charging stations (phones, batteries, medical equipment)

Resilience Measures: - Elevate substations and critical equipment above 100-year flood level (+3m) - Install underground distribution lines in priority areas (more wind-resistant) - Develop solar+battery microgrids for critical facilities (hospital, water, shelters) - Pre-position emergency generators and spare parts in hurricane-resistant storage - Diversify fuel supply (establish secondary fuel depot at elevated inland site)

E. TRANSPORT & ACCESS

Current State (Day 0-3): - **Roads:** RN7 (main coastal road) blocked by debris, flooding, downed power lines; secondary roads impassable - **Bridges:** Several small bridges damaged or washed out (connecting interior to coast) - **Port:** Galion Bay cargo port operational at 30% (damaged cranes, debris in water, fuel dock damaged) - **Airports:** - Princess Juliana (Dutch side, shared): Terminal damaged, runway littered, but structurally intact - Grand Case Airport

(French side): Flooded, debris on runway, terminal damaged - **Vehicles**: ~30% of vehicle fleet damaged (flooded, crushed by debris)

Short-Term Objectives (0-7 days): 1. **Critical Road Clearance** (Day 0-3): - Deploy 15 heavy equipment units (excavators, bulldozers, dump trucks) - Clear RN7 from Marigot to Grand Case to Orient Bay (30 km) - Clear access roads to hospital, port, airport, emergency shelters - Remove downed power lines (coordinate with utility crews) - Target: Main routes passable by Day 3 2. **Airport Restoration** (Day 1-7): - **Grand Case**: Clear runway debris, pump standing water, emergency repairs to terminal - Enable emergency flights (small aircraft, helicopters) by Day 3 - **Princess Juliana**: Clear terminal, restore basic services (fuel, navigation aids) - Enable cargo flights by Day 5 3. **Port Restoration** (Day 1-7): - Clear debris from Galion Bay port (underwater and dock areas) - Repair fuel dock (critical for generator and vehicle fuel) - Restore crane operations (emergency repairs) - Enable cargo ships by Day 5 (target: 60% capacity) 4. **Emergency Transport Fleet** (Day 1-7): - Procure/airlift 10 trucks (5-ton capacity) for logistics - Procure 5 buses for evacuations and shelter transport - Establish fuel distribution system for emergency vehicles

Medium-Term Objectives (8-30 days): 1. **Road Network Restoration** (Week 1-4): - Repair damaged road surfaces (potholes, washouts) on RN7 and main secondary roads - Temporary bridge repairs or detours for damaged bridges (3 priority sites) - Clear and restore 80% of road network by Day 30 2. **Port Expansion** (Week 2-4): - Restore Galion Bay port to 80% capacity - Repair/replace damaged cranes and cargo handling equipment - Dredge harbor entrance if needed (sediment from storm) - Establish efficient cargo clearance system (customs, warehousing) 3. **Airport Full Restoration** (Week 2-4): - Grand Case: Permanent terminal repairs, restore commercial service - Princess Juliana: Full terminal restoration, resume regular flights - Coordinate with airlines for resumption of service 4. **Public Transport** (Week 2-4): - Restore bus service on main routes (Marigot-Grand Case-Quartier d'Orléans) - Subsidize or provide free transport for displaced populations

Resilience Measures: - Elevate coastal road sections prone to storm surge (RN7 in low areas) - Upgrade bridges to withstand 100-year flood events (higher clearance, stronger foundations) - Establish redundant access routes to critical facilities (hospital, port, airport) - Harden port infrastructure (elevate cranes, install breakwaters, dredge for larger vessels) - Upgrade airport drainage systems to prevent flooding

VI. Resilience Project Portfolio

Below are **15 prioritized projects** for implementation within the first month, with specific timelines, locations, and resilience features. These projects are designed to be **concrete actions, not processes**, with clear deliverables.

P1: IMMEDIATE LIFE-SAVING (Days 0-7)

Project 1: Emergency Water Distribution Network

- **Location:** Marigot, Grand Case, Quartier d'Orléans, Cul-de-Sac, Orient Bay (coastal and mid-elevation)
- **Sectors:** Water, Health
- **Timeline:** Day 0-7 (setup), ongoing through Month 1

- **Description:**
 - Deploy 10 water distribution points (schools, churches, community centers)
 - Install 5 emergency water bladders (10,000L each) with purification systems
 - Operate 8 water trucks (5,000L capacity) with daily delivery routes
 - Distribute 100,000 bottles of water (first 72h)
 - **Resilience Features:**
 - Distribution points at elevated sites (>10m) to avoid future flooding
 - Bladders can be rapidly deployed in future emergencies
 - Build community capacity for water point management
 - **Estimated Cost:** \$150,000 (equipment, fuel, personnel)
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Project 2: Hospital Emergency Stabilization & Medical Surge

- **Location:** Louis-Constant Fleming Hospital (Marigot) + 5 mobile clinics island-wide
 - **Sectors:** Health
 - **Timeline:** Day 0–7 (emergency), ongoing improvements through Month 1
 - **Description:**
 - Deploy Emergency Medical Team (20 personnel: surgeons, ER doctors, nurses, anesthetists)
 - Airlift 5 tons of medical supplies (trauma kits, antibiotics, IV fluids, chronic disease meds)
 - Install 200 kW emergency generators with 14-day fuel supply
 - Repair roof and secure critical areas (ER, OR, ICU) with tarps and temporary materials
 - Deploy 5 mobile medical units to underserved areas (Grand Case, Orient Bay, Quartier d'Orléans, Cul-de-Sac, Colombier)
 - **Resilience Features:**
 - Generators elevated above flood level on reinforced platforms
 - Establish medical stockpile in hurricane-resistant container (30-day supply)
 - Train local staff in emergency response protocols
 - **Estimated Cost:** \$800,000 (personnel, supplies, equipment, repairs)
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Project 3: Critical Road & Port Access Clearance

- **Location:** RN7 (Marigot–Grand Case–Orient Bay), port access road, airport access roads
- **Sectors:** Transport, Logistics
- **Timeline:** Day 0–7 (clearance), ongoing maintenance
- **Description:**
- Deploy 15 heavy equipment units (excavators, bulldozers, dump trucks, chainsaws)
- Clear 50 km of priority roads (RN7 and access roads to critical facilities)
- Remove ~10,000 m³ of debris (trees, building materials, vehicles)
- Coordinate with utility crews to remove downed power lines
- Establish 3 debris staging areas (Galion Bay, Quartier d'Orléans, Grand Case)

- **Resilience Features:**
 - Map flood-prone road sections for future elevation/rerouting
 - Pre-position heavy equipment at strategic sites for future events
 - Train local operators in debris management
 - **Estimated Cost:** \$400,000 (equipment rental, fuel, personnel, disposal)
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Project 4: Emergency Shelter Activation & Rapid Repairs

- **Location:** 5 emergency shelters + 500 homes island-wide (priority: Marigot, Grand Case, Quartier d'Orléans)
 - **Sectors:** Shelter, Protection
 - **Timeline:** Day 0–7 (setup and initial repairs)
 - **Description:**
 - Open and equip 5 emergency shelters (capacity: 1,200 people)
 - Provide bedding, food, water, sanitation, hygiene kits
 - Distribute 2,000 tarpaulins (10m x 10m) for roof coverage
 - Deploy 10 shelter repair teams (5 people each) to secure 500 homes
 - Rapid structural assessments (red/yellow/green tagging) for 2,000 homes
 - **Resilience Features:**
 - Shelters in elevated, structurally sound buildings (assessed for future use)
 - Train homeowners in cyclone-resistant tarp installation
 - Document damage for insurance and reconstruction planning
 - **Estimated Cost:** \$300,000 (shelter supplies, tarps, repair materials, personnel)
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Project 5: Fuel Supply Chain Restoration

- **Location:** Galion Bay port fuel depot + distribution network island-wide
 - **Sectors:** Energy, Transport, Logistics
 - **Timeline:** Day 1–7 (restoration), ongoing operations
 - **Description:**
 - Emergency repair of fuel depot (secure tanks, pumps, piping)
 - Airlift/ship 100,000 liters of diesel for first week
 - Establish fuel distribution system (3 trucks, daily deliveries to generators)
 - Ration fuel (priority: hospital, water, shelters, emergency services)
 - Secure fuel depot perimeter and establish 24/7 security
 - **Resilience Features:**
 - Elevate fuel pumps and electrical systems above flood level
 - Establish secondary fuel storage at inland elevated site (Colombier area)
 - Install solar-powered fuel pumps as backup
 - **Estimated Cost:** \$250,000 (repairs, fuel, trucks, security)
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P2: ESSENTIAL SERVICES RESTORATION (Days 8-30)

Project 6: Water Treatment Plant Restoration & Distribution Network Repair

- **Location:** Galion Bay water treatment plant + distribution network (priority: Marigot, Grand Case, Quartier d'Orléans)
 - **Sectors:** Water
 - **Timeline:** Week 1-4 (restoration to 80% capacity by Day 30)
 - **Description:**
 - Repair/replace damaged pumps, motors, electrical systems at treatment plant
 - Install temporary diesel generators (50 kW) for plant operations
 - Repair 5 km of critical water distribution pipes (main trunk lines)
 - Test and chlorinate distribution network
 - Restore service to 50% of population by Day 30
 - **Resilience Features:**
 - Elevate critical plant equipment above 10-year flood level (+2m)
 - Install solar+battery backup power system (design and procurement in Month 1, installation Month 2)
 - Map saltwater intrusion zones and develop protected well network
 - **Estimated Cost:** \$1,200,000 (equipment, materials, labor)
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Project 7: Transitional Shelter Village & Home Repair Program

- **Location:** Quartier d'Orléans and Colombier (mid-elevation sites) + 1,000 homes island-wide
 - **Sectors:** Shelter
 - **Timeline:** Week 2-4 (T-shelter installation), ongoing repairs through Month 1
 - **Description:**
 - Site preparation for 300 transitional shelter units (T-shelters, 18m² each)
 - Procure and install T-shelters with water, sanitation, electricity access
 - Launch home repair program: 20 teams (5 people each) to repair 1,000 homes
 - Distribute shelter repair kits (corrugated metal, lumber, nails, tarps, tools) to 1,000 households
 - Training workshops for homeowners on cyclone-resistant construction (5 sessions, 50 people each)
 - **Resilience Features:**
 - T-shelter sites at elevated locations (>10m) with drainage
 - Home repairs follow "build-back-better" guidelines (wind-resistant roofing, tie-downs)
 - Distribute construction guidelines to all homeowners
 - **Estimated Cost:** \$2,500,000 (T-shelters, materials, labor, training)
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Project 8: Phased Electrical Grid Restoration (Priority Circuits)

- **Location:** Hospital, water plant, airport, port, Marigot downtown, Quartier d'Orléans
 - **Sectors:** Energy
 - **Timeline:** Week 1–4 (restore 30% of island by Day 30)
 - **Description:**
 - **Phase 1 (Week 1–2):** Restore power to critical facilities
 - Install temporary poles and overhead lines (5 km)
 - Repair/replace 3 transformers at key substations
 - Restore power to hospital, water plant, airport, port, police/fire stations
 - **Phase 2 (Week 2–3):** Restore power to Marigot downtown and Quartier d'Orléans
 - Repair distribution network (10 km of lines, 50 poles)
 - Restore service to 1,500 households and businesses
 - **Phase 3 (Week 3–4):** Expand to Grand Case, Cul-de-Sac, Colombier
 - Repair distribution network (15 km of lines, 80 poles)
 - Restore service to 2,000 households
 - **Resilience Features:**
 - Prioritize underground distribution lines in critical areas (hospital, water plant)
 - Elevate substations above flood level (+3m) during repairs
 - Install solar+battery microgrids at hospital and water plant (design in Month 1, installation Month 2–3)
 - **Estimated Cost:** \$3,000,000 (equipment, materials, labor)
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Project 9: Port Infrastructure Restoration & Cargo Capacity Expansion

- **Location:** Galion Bay cargo port
 - **Sectors:** Transport, Logistics
 - **Timeline:** Week 1–4 (restore to 80% capacity by Day 30)
 - **Description:**
 - Clear underwater debris (divers, cranes, barges)
 - Repair damaged dock sections (concrete, pilings)
 - Repair/replace 2 cargo cranes (emergency repairs or temporary cranes)
 - Dredge harbor entrance if sediment accumulation detected
 - Establish efficient cargo clearance system (temporary customs and warehousing)
 - Restore fuel dock to full operations
 - **Resilience Features:**
 - Elevate crane electrical systems above flood level
 - Install breakwaters or floating barriers to reduce wave action
 - Establish backup cargo handling equipment (mobile cranes)
 - **Estimated Cost:** \$1,800,000 (repairs, equipment, dredging, labor)
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Project 10: Airport Restoration & Emergency Airlift Capacity

- **Location:** Grand Case Airport (primary) + Princess Juliana Airport (secondary, Dutch side)
 - **Sectors:** Transport
 - **Timeline:** Week 1-4 (restore commercial service by Day 30)
 - **Description:**
 - **Grand Case:**
 - Clear runway and taxiway debris (heavy equipment, sweepers)
 - Pump standing water from low areas
 - Repair terminal roof and secure building
 - Restore navigation aids and lighting (generators)
 - Enable emergency flights by Day 3, commercial service by Day 21
 - **Princess Juliana** (coordinate with Dutch authorities):
 - Clear terminal debris and restore basic services
 - Restore fuel supply and cargo handling
 - Enable cargo flights by Day 5, passenger flights by Day 14
 - **Resilience Features:**
 - Upgrade drainage systems at Grand Case to prevent flooding
 - Elevate critical navigation equipment above flood level
 - Establish emergency airlift protocols with regional partners
 - **Estimated Cost:** \$900,000 (repairs, equipment, labor; French side only)
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P3: EARLY RESILIENCE & RECOVERY (Days 8-30)

Project 11: Systematic Debris Management & Recycling Program

- **Location:** Island-wide (priority: Marigot, Grand Case, Orient Bay, Quartier d'Orléans)
- **Sectors:** Environment, Shelter, Transport
- **Timeline:** Week 1-4 (process 50,000 m³ by Day 30, ~10% of total)
- **Description:**
 - Establish 3 permanent debris staging and sorting sites (Galion Bay, Quartier d'Orléans, Grand Case)
 - Deploy 20 heavy equipment units (excavators, dump trucks, chippers, compactors)
 - Systematic debris removal from residential areas (street-by-street)
 - Sort debris into categories: recyclable (metal, wood), hazardous (asbestos, chemicals), landfill
 - Process and recycle metals and wood (partner with recycling companies)
 - Safely dispose of hazardous materials (temporary storage for export)
- **Resilience Features:**
 - Establish permanent debris management plan with pre-positioned equipment
 - Train local operators in debris sorting and recycling
 - Map debris hotspots for future rapid response

- **Estimated Cost:** \$1,500,000 (equipment, labor, disposal, recycling)
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Project 12: Community Health & Epidemic Prevention Program

- **Location:** Island-wide (priority: shelters, coastal areas, high-density neighborhoods)
 - **Sectors:** Health, Water, Sanitation
 - **Timeline:** Week 1–4 (ongoing prevention and surveillance)
 - **Description:**
 - Establish disease surveillance system (daily reporting from clinics, hospital, mobile units)
 - Distribute 5,000 hygiene kits (soap, chlorine tablets, ORS, mosquito nets)
 - Deploy vector control teams (spray standing water areas, distribute larvicide)
 - Launch public health campaign (radio spots, posters, community meetings): water treatment, hygiene, vector control
 - Train 50 community health volunteers in disease prevention and surveillance
 - Pre-position epidemic response supplies (cholera/typhoid kits, isolation tents)
 - **Resilience Features:**
 - Build community capacity for disease prevention and surveillance
 - Establish rapid response protocols for outbreaks
 - Map high-risk disease zones (standing water, poor sanitation)
 - **Estimated Cost:** \$200,000 (supplies, training, personnel, campaigns)
-

Project 13: Road Network Restoration & Bridge Repairs

- **Location:** RN7 + secondary roads (RD1, RD2) + 3 priority bridges (connecting interior to coast)
- **Sectors:** Transport
- **Timeline:** Week 1–4 (restore 80% of network by Day 30)
- **Description:**
 - Repair damaged road surfaces (potholes, washouts) on RN7 (30 km)
 - Repair secondary roads (RD1, RD2, others): 40 km total
 - Emergency bridge repairs or temporary detours for 3 damaged bridges:
 - Quartier d'Orléans bridge (over ravine)
 - Cul-de-Sac bridge (over drainage canal)
 - Orient Bay access bridge
 - Install temporary Bailey bridges or culverts if needed
 - Restore 80% of road network by Day 30
- **Resilience Features:**
 - Elevate coastal road sections prone to storm surge (RN7 in low areas) – design in Month 1, construction Month 2+
 - Upgrade bridges to withstand 100-year flood events (higher clearance, stronger foundations) – permanent repairs Month 2+
 - Map flood-prone road sections for future rerouting

- **Estimated Cost:** \$2,200,000 (materials, equipment, labor)
-

Project 14: Distributed Solar+Battery Systems for Critical Facilities

- **Location:** Hospital, water treatment plant, 5 emergency shelters, police/fire stations
 - **Sectors:** Energy, Health, Water
 - **Timeline:** Week 2-4 (design and procurement), installation Month 2-3
 - **Description:**
 - Design solar+battery microgrids for critical facilities (total capacity: 500 kW solar, 1 MWh battery storage)
 - Procure equipment (solar panels, inverters, batteries, mounting systems)
 - Prepare sites (roof reinforcement, ground mounting, electrical integration)
 - Begin installation at hospital and water plant (priority)
 - Target: Hospital and water plant operational on solar+battery by Day 60
 - **Resilience Features:**
 - Facilities can operate independently of grid during outages
 - Solar panels mounted to withstand Category 5 winds
 - Battery storage provides 24h backup power
 - Reduces dependence on diesel fuel
 - **Estimated Cost:** \$1,500,000 (equipment, installation, design; initial procurement in Month 1)
-

Project 15: Coastal Erosion Assessment & Nature-Based Solutions Design

- **Location:** Coastal areas island-wide (priority: Marigot waterfront, Grand Case beach, Orient Bay, Nettle Bay)
- **Sectors:** Environment, Shelter, Transport
- **Timeline:** Week 2-4 (assessment and design), implementation Month 2+
- **Description:**
 - Conduct rapid coastal erosion and damage assessment (satellite imagery, drone surveys, field visits)
 - Map areas with severe erosion, saltwater intrusion, reef damage
 - Design nature-based solutions for coastal protection:
 - Mangrove restoration (Cul-de-Sac, Simpson Bay lagoon)
 - Beach nourishment and dune restoration (Grand Case, Orient Bay)
 - Coral reef restoration (offshore Grand Case, Orient Bay)
 - Living shorelines (Marigot waterfront)
 - Engage community in design (workshops, consultations)
 - Develop implementation plan with timelines and budgets
- **Resilience Features:**
 - Nature-based solutions provide long-term coastal protection
 - Mangroves and reefs reduce storm surge and wave action
 - Restore ecosystem services (fisheries, tourism)

- Build community ownership and stewardship
 - **Estimated Cost:** \$150,000 (assessment, design, consultations; implementation \$2-5M in future phases)
-

VII. Logistics & Operations Plan

A. Supply Chain & Staging

Primary Entry Points: 1. **Galion Bay Port** (main cargo port, French side): - **Status:** Operational at 30% Day 0; target 60% by Day 5, 80% by Day 30 - **Function:** Bulk cargo (fuel, food, water, construction materials, heavy equipment) - **Capacity:** 2-3 cargo ships per week by Day 30 - **Staging Area:** 5-hectare site adjacent to port for cargo sorting and temporary storage 2. **Grand Case Airport** (French side): - **Status:** Emergency flights by Day 3; commercial service by Day 21 - **Function:** Humanitarian airlift (medical supplies, personnel, high-priority cargo) - **Capacity:** 5-10 flights per day (small aircraft, helicopters) 3. **Princess Juliana Airport** (Dutch side, shared): - **Status:** Cargo flights by Day 5; passenger flights by Day 14 - **Function:** Large cargo aircraft, passenger evacuations/arrivals, humanitarian airlift - **Capacity:** 10-20 flights per day by Day 14

Warehousing & Distribution: - **Central Warehouse** (Galion Bay area, near port): - 2,000 m² temporary warehouse (tents or containers) - Function: Receiving, sorting, and distributing relief supplies - Staffing: 20 logistics personnel (warehouse managers, forklift operators, inventory clerks) - Operating hours: 24/7 - **Forward Distribution Points** (5 sites): - Marigot (downtown): 500 m² - serves western coastal areas - Grand Case (airport area): 300 m² - serves northern coastal areas - Quartier d'Orléans (school): 400 m² - serves mid-elevation residential areas - Cul-de-Sac (community center): 300 m² - serves southern lowlands - Colombier (church): 200 m² - serves inland areas - Function: Last-mile distribution to communities and households - Staffing: 3-5 personnel per site

Transport Fleet: - **Trucks:** 10 x 5-ton capacity (for cargo distribution from port/airport to warehouses and distribution points) - **Light vehicles:** 20 x 4WD pickups (for mobile clinics, assessment teams, rapid response) - **Buses:** 5 x 50-passenger capacity (for evacuations, shelter transport, community access) - **Fuel tankers:** 3 x 5,000-liter capacity (for fuel distribution to generators, vehicles, equipment) - **Heavy equipment:** 15 units (excavators, bulldozers, dump trucks for debris clearance and construction)

Fuel Supply: - **Demand:** ~10,000 liters/day (Week 1), ~15,000 liters/day (Weeks 2-4) -

Supply: Fuel ships to Galion Bay port every 3-5 days (50,000-100,000 liters per shipment) -

Distribution: From port depot to generators, vehicles, equipment via tanker trucks -

Rationing: Priority system (P1: hospital, water, shelters; P2: emergency services, logistics; P3: reconstruction)

B. Access Corridors & Detours

Primary Access Corridors (Week 1): 1. **RN7 Coastal Route** (Marigot-Grand Case-Orient Bay): - **Status:** Cleared by Day 3; passable for all vehicles - **Function:** Main artery for humanitarian logistics - **Constraints:** Flooding in low sections during heavy rain; temporary repairs only 2. **Port Access Road** (RN7 to Galion Bay): - **Status:** Cleared by Day 2; passable for heavy trucks - **Function:** Critical for cargo offloading and fuel supply 3. **Airport Access**

Roads: - Grand Case: Cleared by Day 2 - Princess Juliana (via Dutch side): Cleared by Day 3 (coordinate with Dutch authorities)

Secondary Access Corridors (Week 2+): - **RD1 Interior Route** (Marigot–Quartier d'Orléans–Oyster Pond): - **Status:** Cleared by Day 7; passable for light vehicles - **Function:** Access to mid-elevation residential areas and eastern coast - **Constraints:** Landslides, narrow sections - **RD2 Southern Route** (Marigot–Cul-de-Sac–Oyster Pond): - **Status:** Cleared by Day 10; passable for light vehicles - **Function:** Access to southern lowlands and eastern coast - **Constraints:** Flooding, bridge damage

Detours & Temporary Solutions: - **Quartier d'Orléans Bridge** (over ravine): Damaged; temporary detour via RD1 alternate route (adds 10 minutes) - **Cul-de-Sac Bridge** (over drainage canal): Damaged; temporary culvert installed by Day 14 - **Orient Bay Access Bridge:** Damaged; temporary Bailey bridge installed by Day 21

Access to Remote Communities: - **Grand Case Village** (coastal): Accessible via RN7 by Day 3 - **Orient Bay** (coastal): Accessible via RN7 by Day 3 - **Colombier** (inland, elevated): Accessible via RD1 by Day 7 - **Oyster Pond** (eastern coast): Accessible via RD1 or RD2 by Day 10 - **Baie Nettle** (western coast): Accessible via RN7 by Day 3

C. Weather & Seasonal Considerations

Hurricane Season (June 1 – November 30): - **Peak:** August–October - **Irma Timing:** September 2017 (peak season) - **Immediate Risk:** Additional tropical storms or hurricanes in September–October 2017 - **Mitigation:** - Monitor weather forecasts daily (NOAA, Météo France) - Establish rapid shelter-in-place protocols for workers and communities - Pre-position emergency supplies (tarps, food, water) at distribution points - Secure temporary structures (T-shelters, tents) against wind

Rainy Season (May–November): - **Peak:** September–November (coincides with hurricane season) - **Impacts:** Flash flooding, landslides, road washouts, standing water (disease risk) - **Mitigation:** - Prioritize drainage clearance (culverts, ditches, storm drains) - Monitor flood-prone areas (Cul-de-Sac, low sections of RN7) - Establish flood early warning system (rain gauges, community observers)

Dry Season (December–April): - **Conditions:** Lower rainfall, calmer seas, cooler temperatures - **Opportunities:** Ideal for construction and reconstruction (less weather disruption) - **Note:** Irma recovery will extend into dry season (December 2017+), allowing accelerated work

D. Security & Coordination

Security Situation (Week 1): - **Risks:** Looting, fuel theft, tensions over scarce resources - **Response:** - Deploy/reinforce police and gendarmerie (French security forces) - Establish 24/7 security at port, airport, fuel depot, warehouses - Community policing in shelters and distribution points - Curfew if needed (coordinate with Préfecture)

Coordination Mechanisms: 1. **Emergency Operations Center (EOC)** (Préfecture, Marigot): - **Function:** Central coordination of all response activities - **Participants:** Préfet, mayors, sector leads (health, water, energy, transport, shelter), military, NGOs, UN agencies - **Meetings:** Daily at 8:00 AM and 6:00 PM (Week 1), then daily at 9:00 AM (Week 2+) 2. **Sector Working Groups** (co-located at EOC or virtual): - **Health:** Led by ARS (Agence Régionale de Santé) - **Water/Sanitation:** Led by SAUR or water authority - **Energy:** Led by EDF (Électricité

de France) or local utility - **Shelter**: Led by social services and NGOs (Red Cross, etc.) - **Transport/Logistics**: Led by Préfecture logistics unit - **Meetings**: Daily or as needed 3. **Community Liaison**: - Engage community leaders (mayors, neighborhood associations, church leaders) - Establish community feedback mechanisms (hotlines, community meetings, suggestion boxes) - Deploy community mobilizers (local volunteers) to support distribution and communication 4. **International Coordination**: - Coordinate with Dutch authorities (Sint Maarten side) on shared infrastructure (airport, port, utilities) - Engage UN agencies (OCHA, UNDP, UNICEF, WFP) for technical support and funding - Coordinate with regional partners (CDEMA, CARICOM) for mutual aid

VIII. Financial & Partnership Strategy

A. Cost Estimates (Month 1)

Category	Week 1 (Days 0-7)	Weeks 2-4 (Days 8-30)	Total Month 1
Water & Sanitation	\$150,000	\$1,200,000	\$1,350,000
Health	\$800,000	\$400,000	\$1,200,000
Shelter & Debris	\$600,000	\$4,000,000	\$4,600,000
Energy	\$250,000	\$4,500,000	\$4,750,000
Transport & Logistics	\$400,000	\$4,900,000	\$5,300,000
Other (security, coordination, community programs)	\$100,000	\$300,000	\$400,000
TOTAL	\$2,300,000	\$15,300,000	\$17,600,000

Notes: - Estimates are rough order of magnitude; actual costs will vary based on procurement, labor rates, and availability - Does not include long-term reconstruction (Months 2+), which will require hundreds of millions - Assumes mix of local procurement, regional imports, and international airlift

B. Funding Sources

Immediate (Week 1): 1. **French Government Emergency Fund**: - Activate national disaster response budget (Fonds d'Urgence) - Target: €2-5 million for first week - Covers: Emergency supplies, personnel deployment, airlift, generators 2. **Territorial Budget** (Collectivité de Saint-Martin): - Redirect available funds to emergency response - Target: €500,000-1 million - Covers: Local contracts, fuel, immediate repairs 3. **Humanitarian Agencies** (UN, Red Cross, NGOs): - Emergency appeals (CERF, IFRC, bilateral donors) - Target: \$1-3 million for first week - Covers: Mobile clinics, shelter kits, water distribution, logistics

Medium-Term (Weeks 2-4): 1. **French Government Reconstruction Fund**: - Establish dedicated Saint-Martin reconstruction program - Target: €50-100 million for first month (part of larger multi-year package) - Covers: Infrastructure restoration (water, energy, transport), shelter, health 2. **European Union**: - EU Civil Protection Mechanism (in-kind support:

equipment, personnel) - EU Solidarity Fund (financial support for reconstruction) - Target: €10–20 million for first month 3. **International Financial Institutions:** - World Bank, Inter-American Development Bank (IDB), Caribbean Development Bank (CDB) - Emergency loans and grants - Target: \$20–50 million for first month (part of larger reconstruction package) 4. **Private Sector & Insurance:** - Insurance payouts for damaged businesses and homes (will take weeks to months) - Corporate donations (telecom, energy, construction companies) - Target: \$5–10 million for first month 5. **Diaspora & Crowdfunding:** - Saint-Martin diaspora in France, US, Netherlands - Online crowdfunding campaigns - Target: \$1–2 million for first month

C. Procurement Strategy

Week 1 (Emergency): - **Method:** Direct procurement, no competitive bidding (emergency waiver) - **Sources:** - Local suppliers (for available goods and services) - Regional suppliers (Guadeloupe, Martinique, Puerto Rico, US Virgin Islands) - International airlift (for critical supplies not available locally) - **Priorities:** Speed over cost; ensure quality and accountability

Weeks 2–4 (Transition): - **Method:** Framework agreements with pre-qualified suppliers (faster than full tenders) - **Sources:** - Expand regional procurement (Caribbean, US, Europe) - Establish long-term agreements for reconstruction materials (cement, steel, lumber, roofing) - **Priorities:** Balance speed, cost, and quality; build local capacity

Month 2+ (Reconstruction): - **Method:** Competitive tenders for large contracts (roads, bridges, energy, water infrastructure) - **Sources:** International and regional contractors - **Priorities:** Cost-effectiveness, quality, sustainability, local employment

D. Partnership & Coordination

Key Partners: 1. **French Government:** - Préfecture (lead coordination) - Ministries: Interior, Overseas Territories, Health, Ecology, Armed Forces - Agencies: ARS (health), EDF (energy), SAUR (water), Météo France (weather) 2. **Territorial Government** (Collectivité de Saint-Martin): - President, Territorial Council - Technical services (public works, social services, urbanism) 3. **Dutch Authorities** (Sint Maarten): - Governor, Prime Minister - Coordination on shared infrastructure (airport, port, utilities) 4. **UN Agencies:** - OCHA (coordination), UNDP (recovery planning), UNICEF (child protection, WASH), WFP (logistics), WHO (health) 5. **NGOs:** - Red Cross (shelter, health, WASH) - Oxfam, CARE, Mercy Corps (WASH, livelihoods) - Local NGOs and community organizations 6. **Private Sector:** - Utilities (EDF, SAUR, telecom companies) - Construction companies (local and regional) - Hospitality sector (hotels for temporary housing, logistics) 7. **Regional Organizations:** - CDEMA (Caribbean Disaster Emergency Management Agency): Technical support, equipment - CARICOM (Caribbean Community): Political support, resource mobilization

Coordination Mechanisms: - **Daily EOC meetings** (Week 1), then as needed - **Sector working groups** (ongoing) - **Community consultations** (weekly in each neighborhood) - **Donor coordination meetings** (weekly, led by Préfecture and UN)

IX. Risk Register

Risk Category	Specific Risk	Likelihood	Impact	Mitigation
Operational		Medium		

Risk Category	Specific Risk	Likelihood	Impact	Mitigation
	Additional hurricane/ tropical storm in Sept- Oct 2017		Very High	Weather monitoring, rapid shelter-in-place protocols, pre- positioned supplies
Operational	Fuel shortage (supply chain disruption)	Medium	High	Diversify fuel sources, establish secondary storage, ration fuel, solar backup
Operational	Heavy equipment shortage (high regional demand post-Irma)	High	High	Pre-position equipment, long- term rental agreements, import from outside region
Operational	Skilled labor shortage (electricians, plumbers, construction workers)	High	Medium	Deploy personnel from France/ EU, train local workers, use military engineers
Health	Waterborne disease outbreak (cholera, typhoid)	Medium	High	Water quality monitoring, hygiene promotion, pre- positioned medical supplies, rapid response teams
Health	Vector-borne disease outbreak (dengue, Zika)	Medium	Medium	Vector control (spraying, larvicide), mosquito nets, public health campaigns
Environmental	Coastal flooding from heavy rain (damaged drainage systems)	High	Medium	Prioritize drainage clearance, flood early warning, avoid construction in flood zones
Environmental	Landslides (saturated soil, damaged vegetation)	Medium	Medium	Avoid construction on steep slopes, stabilize critical slopes, monitor weather
Environmental	Hazardous material release (fuel, chemicals from damaged facilities)	Low	High	Rapid containment, safe disposal, environmental assessment
Social	Community tensions over resource distribution	Medium	Medium	Transparent distribution system, community engagement, grievance mechanisms
Social	Looting and insecurity	Medium	Medium	Police/gendarmerie presence, secure warehouses and fuel, community policing
Governance	Coordination gaps between French/Dutch sides	Medium	Medium	Joint coordination meetings, shared information systems, mutual aid agreements
Governance	Slow decision-making (bureaucracy)	Medium	Medium	Emergency waivers, streamlined approval processes, delegate authority to field teams
Financial		High	High	

Risk Category	Specific Risk	Likelihood	Impact	Mitigation
	Funding delays (disbursement bottlenecks)			Advance funding from emergency reserves, flexible budget reallocations, donor coordination
Financial	Cost overruns (high demand, limited supply)	High	Medium	Contingency budgets (20%), framework agreements, cost monitoring

X. Strategic Roadmap

Phase 0: IMMEDIATE RESPONSE (Days 0-7)

Objective: Save lives, stabilize critical systems, enable humanitarian access

Key Milestones: - Day 0: EOC activated, emergency shelters opened, water distribution begins - Day 1: Hospital stabilized, mobile clinics deployed, fuel supply restored - Day 2: Port access cleared, airport emergency operations begin - Day 3: RN7 cleared, emergency flights operational, 1,200 people sheltered - Day 5: Port operational at 60%, cargo flights operational, water to 50% of population - Day 7: 500 homes secured with tarps, 80% of road network cleared, disease surveillance active

Success Criteria: - Zero preventable deaths from lack of water, shelter, or medical care - All critical facilities (hospital, water plant, port, airport) operational - Humanitarian supply chain functional - No disease outbreaks

Phase 1: ESSENTIAL SERVICES RESTORATION (Days 8-30)

Objective: Restore water, power, shelter, and transport to majority of population; prevent epidemics; begin early reconstruction

Key Milestones: - Day 10: Water plant at 60% capacity, power to critical facilities, 80% of roads passable - Day 14: 300 T-shelters under construction, airport commercial service resumed, port at 70% capacity - Day 21: Water to 50% of population, power to 15% of island, 1,000 homes repaired - Day 30: Water to 70% of population, power to 30% of island, 1,500 homes repaired, 300 T-shelters occupied

Success Criteria: - 70% of population with access to safe water - 30% of population with electricity - 2,000 households in safe shelter (repaired homes or T-shelters) - No major disease outbreaks - Port and airport fully operational - 80% of road network restored

Phase 2: EARLY RECOVERY & RESILIENCE (Months 2-6)

Objective: Complete restoration of basic services, accelerate reconstruction, implement resilience measures

Key Activities: - **Water:** Full restoration of treatment plant and distribution network; solar backup installed - **Energy:** Restore power to 80% of island; install solar+battery microgrids at

critical facilities - **Shelter**: Complete T-shelter program (300 units); repair 5,000 homes; begin permanent housing reconstruction - **Transport**: Permanent road and bridge repairs; upgrade coastal roads; restore public transport - **Health**: Full hospital restoration; reopen all health centers; mental health program expansion - **Debris**: Process 80% of debris (400,000 m³); establish permanent management system - **Coastal Protection**: Implement nature-based solutions (mangrove restoration, beach nourishment, reef restoration)

Success Criteria: - 100% of population with access to safe water - 80% of population with electricity - 5,000+ households in safe, permanent shelter - All critical infrastructure restored and resilient - Coastal protection measures underway

Phase 3: LONG-TERM RECONSTRUCTION & RESILIENCE (Months 7-24+)

Objective: Build back better, enhance long-term resilience, restore livelihoods and economy

Key Activities: - **Infrastructure**: Complete rebuild of energy grid (underground lines, elevated substations, renewable energy) - **Coastal Protection**: Complete nature-based solutions; establish marine protected areas - **Housing**: Complete reconstruction of 4,000 destroyed homes (cyclone-resistant, elevated in flood zones) - **Economic Recovery**: Restore tourism sector (hotels, marinas, beaches); support small businesses - **Governance**: Update building codes, land-use planning, disaster risk reduction plans - **Capacity Building**: Train local workforce in resilient construction, emergency management, environmental stewardship

Success Criteria: - All infrastructure rebuilt to higher resilience standards - Zero households in temporary shelter - Tourism and economy restored to pre-Irma levels or better - Community resilience and preparedness significantly enhanced - Saint-Martin recognized as model for island resilience

CONCLUSION & NEXT STEPS

This plan provides a comprehensive, prioritized roadmap for Saint-Martin's recovery from Hurricane Irma, focusing on the critical first week and first month. The **15 concrete projects** outlined in Section VI are designed to be implemented immediately, with clear timelines, locations, and resilience features.

Immediate Actions (Next 24-48 Hours):

1. **Activate EOC** and convene first coordination meeting with all partners
2. **Deploy emergency water distribution** (trucks, bladders, bottled water)
3. **Stabilize hospital** (medical team, supplies, generators, roof repair)
4. **Clear critical roads** (RN7, port access, airport access)
5. **Open emergency shelters** and begin rapid home repairs (tarps)
6. **Restore fuel supply** (repair depot, airlift diesel)

Priority for Week 1:

- Focus on **Projects 1-5** (life-saving and access)
- Establish logistics system (warehouses, transport fleet, fuel distribution)

- Begin damage assessments for medium-term projects

Priority for Weeks 2-4:

- Implement **Projects 6-15** (services restoration and early resilience)
- Scale up shelter, water, energy, and transport programs
- Engage communities in planning and implementation

Critical Success Factor:

Integration of local knowledge and community priorities is essential. The CSV file mentioned in your request (prioritizing areas in Saint-Martin) should be uploaded to the RAG system to refine this plan with specific data on: - Population density and vulnerability by neighborhood - Infrastructure criticality scores - Flood/storm surge risk zones - Access constraints - Community priorities

Once that data is available, I can provide a **more granular, data-driven prioritization** of projects and areas, ensuring that the most vulnerable and critical locations receive attention first.

This plan is designed to be actionable, measurable, and resilient—moving Saint-Martin from emergency response to long-term recovery within the first month, while laying the foundation for a more resilient future.